

AERIS Expert Review Packet

Anomaly

Source / metric	tceq / no2
Observed / expected	33.5 / 8.84444
Time	2026-06-01 20:00 UTC
Location	29.8144, -95.3878
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	humidity	percent	9 / 478	43.99 to 100; mean 79.5181	52.85 at 2026-06-01 19:55Z; dt 5 min
asos	temperature	degC	9 / 478	21.6667 to 34; mean 27.4992	34 at 2026-06-01 19:55Z; dt 5 min
asos	wind_direction	degree	9 / 482	0 to 360; mean 96.3693	100 at 2026-06-01 19:55Z; dt 5 min
asos	wind_speed	m/s	9 / 482	0 to 8.2311; mean 2.3246	3.60111 at 2026-06-01 19:55Z; dt 5 min
noaa_gfs	gh_500	m	11 / 132	5877.66 to 5901.83; mean 5889.36	5900.97 at 2026-06-01 18:00Z; dt 120 min
noaa_gfs	pbl_height	m	11 / 132	14.9006 to 2020.13; mean 648.061	1863.81 at 2026-06-01 18:00Z; dt 120 min
noaa_gfs	precipitable_water	mm	11 / 132	42.8617 to 53.6474; mean 47.8189	47.7531 at 2026-06-01 18:00Z; dt 120 min
noaa_gfs	surface_pressure	hPa	11 / 132	1004.18 to 1013.92; mean 1010.38	1010.44 at 2026-06-01 18:00Z; dt 120 min
noaa_gfs	t_850	degC	11 / 132	16.5256 to 19.6844; mean 18.1622	17.3775 at 2026-06-01 18:00Z; dt 120 min
noaa_gfs	u_10m	m/s	11 / 132	-3.67282 to 0.747175; mean -1.4459	-0.772268 at 2026-06-01 18:00Z; dt 120 min
noaa_gfs	v_10m	m/s	11 / 132	-1.54819 to 3.78564; mean 1.206	0.676589 at 2026-06-01 18:00Z; dt 120 min
openaq	ozone	ppm	13 / 651	-0.002 to 0.094; mean 0.0213	0.052 at 2026-06-01 20:00Z; dt 0 min
openaq	pm10	ug/m3	2 / 102	9 to 110; mean 27.9314	30 at 2026-06-01 20:00Z; dt 0 min
openaq	pm25	ug/m3	28 / 2376	-4 to 34.5; mean 5.7312	12.4 at 2026-06-01 20:00Z; dt 0 min
openweather	cloud_cover	percent	5 / 360	0 to 100; mean 45.3083	40 at 2026-06-01 19:57Z; dt 2.4 min
openweather	humidity	percent	5 / 360	57 to 95; mean 78.85	57 at 2026-06-01 19:57Z; dt 2.4 min
openweather	precipitation	mm/h	5 / 360	0 to 6.48; mean 0.1034	0 at 2026-06-01 19:57Z; dt 2.4 min
openweather	pressure	hPa	5 / 360	1011 to 1016; mean 1013.48	1013 at 2026-06-01 19:57Z; dt 2.4 min
openweather	temperature	degC	5 / 360	22.42 to 34.52; mean 28.2797	33.89 at 2026-06-01 19:57Z; dt 2.4 min
openweather	wind_direction	degree	5 / 360	0 to 360; mean 123.653	80 at 2026-06-01 19:57Z; dt 2.4 min
openweather	wind_speed	m/s	5 / 360	0 to 7.2; mean 2.3875	3.09 at 2026-06-01 19:57Z; dt 2.4 min
purpleair	pm25	ug/m3	65 / 3688	0 to 58; mean 6.0837	9.6 at 2026-06-01 20:00Z; dt 0 min

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
sentinel5p	s5p_aer_ai_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-06-01 20:07Z; dt 7.2 min
sentinel5p	s5p_ch4_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-06-01 20:07Z; dt 7.2 min
sentinel5p	s5p_co_column	mol/m^2	4 / 4	0.028772 to 0.0302015; mean 0.0295	0.0302015 at 2026-06-01 20:07Z; dt 7.2 min
sentinel5p	s5p_co_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-06-01 20:07Z; dt 7.2 min
sentinel5p	s5p_hcho_column	mol/m^2	5 / 5	0.000117197 to 0.000169173; mean 0.0001	0.000162615 at 2026-06-01 20:07Z; dt 7.2 min
sentinel5p	s5p_hcho_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-06-01 20:07Z; dt 7.2 min
sentinel5p	s5p_no2_column	mol/m^2	6 / 6	1.483e-05 to 4.38402e-05; mean 0	4.38402e-05 at 2026-06-01 20:07Z; dt 7.2 min
sentinel5p	s5p_no2_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-06-01 20:07Z; dt 7.2 min
sentinel5p	s5p_o3_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-06-01 20:07Z; dt 7.2 min
sentinel5p	s5p_so2_column	mol/m^2	5 / 5	-9.16616e-06 to 0.000116468; mean 0	8.81783e-06 at 2026-06-01 20:07Z; dt 7.2 min
sentinel5p	s5p_so2_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-06-01 20:07Z; dt 7.2 min
tceq	co	ppm	3 / 215	0 to 0.9; mean 0.2098	0.4 at 2026-06-01 20:00Z; dt 0 min
tceq	no2	ppb	13 / 896	-2.6 to 34.4; mean 4.985	33.5 at 2026-06-01 20:00Z; dt 0 min
tceq	so2	ppb	3 / 212	-0.5 to 1.6; mean -0.0613	0 at 2026-06-01 20:00Z; dt 0 min

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	humidity	06-01 00Z 82.65, 06Z 95.36, 12Z 70.61, 18Z 60.8, 06-02 00Z 84.01, 06Z 94.18, 12Z 76.34, 18Z 66.96, 06-03 00Z 84.58, 06Z 88.53
asos	temperature	06-01 00Z 26.77, 06Z 24.39, 12Z 29.37, 18Z 31.67, 06-02 00Z 27.27, 06Z 25.14, 12Z 28.78, 18Z 29.95, 06-03 00Z 24.49, 06Z 24.07
asos	wind_direction	06-01 00Z 133.9, 06Z 21.32, 12Z 80.57, 18Z 135.6, 06-02 00Z 129.4, 06Z 52.16, 12Z 98.15, 18Z 108.9, 06-03 00Z 77.02, 06Z 193.1
asos	wind_speed	06-01 00Z 2.541, 06Z 0.3106, 12Z 1.281, 18Z 3.782, 06-02 00Z 1.998, 06Z 0.5851, 12Z 1.982, 18Z 3.931, 06-03 00Z 4.389, 06Z 3.247
noaa_gfs	gh_500	05-31 12Z 5880, 18Z 5896, 06-01 00Z 5891, 06Z 5900, 12Z 5885, 18Z 5900, 06-02 00Z 5888, 06Z 5898, 12Z 5881, 18Z 5894, 06-03 00Z 5880, 06Z 5880
noaa_gfs	pbl_height	05-31 12Z 148.8, 18Z 1584, 06-01 00Z 732.2, 06Z 225.8, 12Z 67.97, 18Z 1700, 06-02 00Z 564.7, 06Z 86.98, 12Z 42.25, 18Z 1654, 06-03 00Z 748.6, 06Z 221.8
noaa_gfs	precipitable_water	05-31 12Z 43.83, 18Z 45.68, 06-01 00Z 47.21, 06Z 46.22, 12Z 46.21, 18Z 48.98, 06-02 00Z 51.86, 06Z 49.79, 12Z 50.16, 18Z 48.67, 06-03 00Z 46.9, 06Z 48.34
noaa_gfs	surface_pressure	05-31 12Z 1010, 18Z 1010, 06-01 00Z 1008, 06Z 1011, 12Z 1011, 18Z 1011, 06-02 00Z 1009, 06Z 1011, 12Z 1011, 18Z 1012, 06-03 00Z 1010, 06Z 1012
noaa_gfs	t_850	05-31 12Z 18.83, 18Z 16.91, 06-01 00Z 18.45, 06Z 18.58, 12Z 17.84, 18Z 17.33, 06-02 00Z 18.76, 06Z 18.54, 12Z 17.59, 18Z 16.97, 06-03 00Z 19.2, 06Z 18.95
noaa_gfs	u_10m	05-31 12Z -0.6771, 18Z -1.243, 06-01 00Z -2.565, 06Z -0.7446, 12Z -0.4489, 18Z -1.458, 06-02 00Z -2.194, 06Z -0.3987, 12Z -0.483, 18Z -1.96, 06-03 00Z -2.718, 06Z -2.461

Source	Metric	Values
noaa_gfs	v_10m	05-31 12Z 1.595, 18Z 2.177, 06-01 00Z 2.902, 06Z 1.967, 12Z 0.7749, 18Z 0.7166, 06-02 00Z 2.65, 06Z 1.596, 12Z -1.005, 18Z -1.215, 06-03 00Z 1.894, 06Z 0.4182
openaq	ozone	06-01 06Z 0.009731, 12Z 0.01646, 18Z 0.04762, 06-02 00Z 0.01781, 06Z 0.007731, 12Z 0.01115, 18Z 0.03371, 06-03 00Z 0.02384, 06Z 0.02531
openaq	pm10	06-01 06Z 31.42, 12Z 44.58, 18Z 30.83, 06-02 00Z 25.17, 06Z 22.08, 12Z 31.17, 18Z 28.42, 06-03 00Z 17.5, 06Z 12.5
openaq	pm25	06-01 06Z 6.451, 12Z 5.868, 18Z 6.815, 06-02 00Z 5.303, 06Z 4.973, 12Z 6.72, 18Z 5.216, 06-03 00Z 5.146, 06Z 3.379
openweather	cloud_cover	05-31 06Z 35.1, 12Z 61.93, 18Z 55.3, 06-01 00Z 20.7, 06Z 41.5, 12Z 43.47, 18Z 44.27, 06-02 00Z 20.8, 06Z 24.37, 12Z 54.9, 18Z 83.7, 06-03 00Z 38.03, 06Z 94
openweather	humidity	05-31 06Z 90.8, 12Z 74.57, 18Z 64.47, 06-01 00Z 81.03, 06Z 92.27, 12Z 74.93, 18Z 64.9, 06-02 00Z 80, 06Z 91.3, 12Z 78.7, 18Z 69.03, 06-03 00Z 85.87, 06Z 85.8
openweather	precipitation	05-31 06Z 0, 12Z 0, 18Z 0, 06-01 00Z 0, 06Z 0.005, 12Z 0, 18Z 0.03, 06-02 00Z 0, 06Z 0, 12Z 0, 18Z 1.003, 06-03 00Z 0.2023, 06Z 0
openweather	pressure	05-31 06Z 1012, 12Z 1014, 18Z 1011, 06-01 00Z 1013, 06Z 1014, 12Z 1014, 18Z 1012, 06-02 00Z 1013, 06Z 1014, 12Z 1015, 18Z 1013, 06-03 00Z 1015, 06Z 1015
openweather	temperature	05-31 06Z 24.68, 12Z 29.25, 18Z 32.05, 06-01 00Z 27.42, 06Z 24.92, 12Z 29.78, 18Z 32.53, 06-02 00Z 28.22, 06Z 25.73, 12Z 29.28, 18Z 31.44, 06-03 00Z 24.26, 06Z 24.04
openweather	wind_direction	05-31 06Z 111.9, 12Z 157.6, 18Z 142.1, 06-01 00Z 151, 06Z 79.53, 12Z 122.5, 18Z 133.8, 06-02 00Z 166.2, 06Z 68.2, 12Z 121.1, 18Z 129.9, 06-03 00Z 91.73, 06Z 136.5
openweather	wind_speed	05-31 06Z 1.367, 12Z 2.946, 18Z 4.11, 06-01 00Z 3.397, 06Z 1.041, 12Z 1.079, 18Z 3.923, 06-02 00Z 3.192, 06Z 0.8987, 12Z 1.722, 18Z 2.616, 06-03 00Z 2.312, 06Z 1.503
purpleair	pm25	06-01 00Z 5.865, 06Z 7.763, 12Z 5.952, 18Z 6.977, 06-02 00Z 5.427, 06Z 5.803, 12Z 6.927, 18Z 4.514, 06-03 00Z 5.967, 06Z 5.24
sentinel5p	s5p_aer_ai_granule_available	05-31 18Z 1, 06-01 18Z 1, 06-02 18Z 1
sentinel5p	s5p_ch4_granule_available	05-31 18Z 1, 06-01 18Z 1, 06-02 18Z 1
sentinel5p	s5p_co_column	06-01 18Z 0.03012, 06-02 18Z 0.02882
sentinel5p	s5p_co_granule_available	05-31 18Z 1, 06-01 18Z 1, 06-02 18Z 1
sentinel5p	s5p_hcho_column	05-31 18Z 0.0001186, 06-01 18Z 0.0001645, 06-02 18Z 0.0001692
sentinel5p	s5p_hcho_granule_available	05-31 18Z 1, 06-01 18Z 1, 06-02 18Z 1
sentinel5p	s5p_no2_column	05-31 18Z 1.667e-05, 06-01 18Z 4.08e-05, 06-02 18Z 4.07e-05
sentinel5p	s5p_no2_granule_available	05-31 18Z 1, 06-01 18Z 1, 06-02 18Z 1
sentinel5p	s5p_o3_granule_available	05-31 18Z 1, 06-01 18Z 1, 06-02 18Z 1
sentinel5p	s5p_so2_column	05-31 18Z 0.0001165, 06-01 18Z 1.316e-05, 06-02 18Z -8.155e-06
sentinel5p	s5p_so2_granule_available	05-31 18Z 1, 06-01 18Z 1, 06-02 18Z 1
tceq	co	05-31 06Z 0.175, 12Z 0.2167, 18Z 0.22, 06-01 00Z 0.25, 06Z 0.2833, 12Z 0.2529, 18Z 0.2278, 06-02 00Z 0.2, 06Z 0.2333, 12Z 0.2056, 18Z 0.1667, 06-03 00Z 0.1389, 06Z 0.08889
tceq	no2	05-31 06Z 3.434, 12Z 3.055, 18Z 2.811, 06-01 00Z 3.459, 06Z 5.427, 12Z 6.123, 18Z 5.614, 06-02 00Z 4.532, 06Z 5.082, 12Z 7.521, 18Z 6.109, 06-03 00Z 5.897, 06Z 5.074
tceq	so2	05-31 06Z -0.27, 12Z -0.2467, 18Z -0.1556, 06-01 00Z -0.1556, 06Z -0.1778, 12Z 0.1889, 18Z 0.08333, 06-02 00Z -0.1444, 06Z -0.1294, 12Z -0.01176, 18Z 0.1556, 06-03 00Z -0.01667, 06Z -0.02222

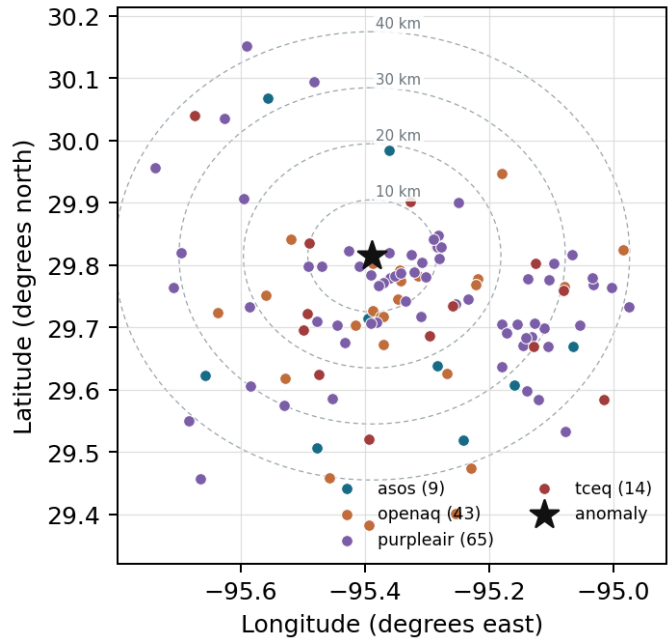
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

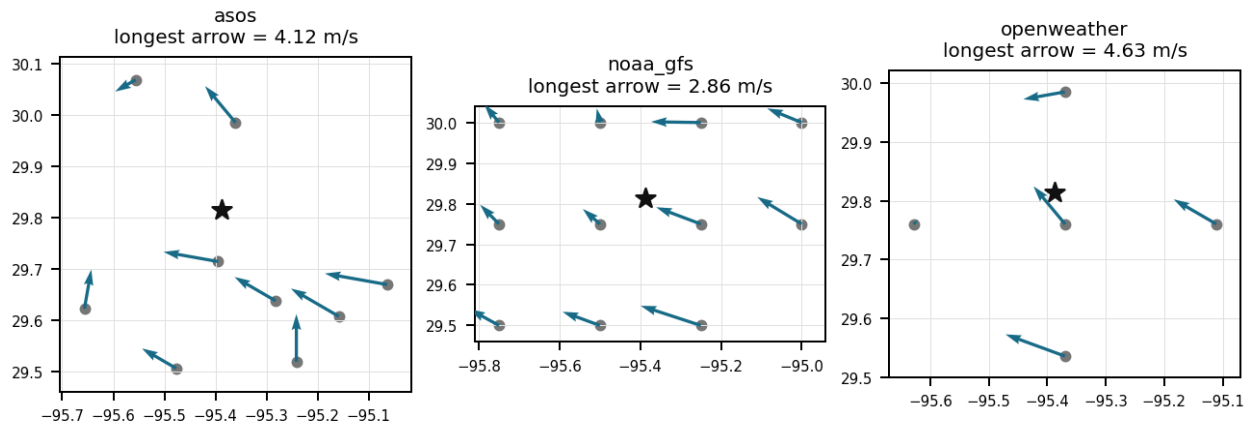
Source	Metric	Values
asos	humidity	MCJ (11.184 km) 77.52, IAH (19.084 km) 80.65, HOU (22.147 km) 76.97, EFD (31.936 km) 78.99, DWH (32.534 km) 85.16, SGR (33.607 km) 79.84, T41 (35.177 km) 81.08, AXH (35.344 km) 76.38, +1 more
asos	temperature	MCJ (11.184 km) 27.98, IAH (19.084 km) 27.37, HOU (22.147 km) 28.02, EFD (31.936 km) 27.92, DWH (32.534 km) 26.02, SGR (33.607 km) 27.94, T41 (35.177 km) 27.45, AXH (35.344 km) 27.46, +1 more
asos	wind_direction	MCJ (11.184 km) 108.5, IAH (19.084 km) 107.3, HOU (22.147 km) 115, EFD (31.936 km) 90.82, DWH (32.534 km) 77.5, SGR (33.607 km) 115.5, T41 (35.177 km) 108.2, AXH (35.344 km) 68.04, +1 more
asos	wind_speed	MCJ (11.184 km) 3.698, IAH (19.084 km) 2.37, HOU (22.147 km) 2.747, EFD (31.936 km) 2.436, DWH (32.534 km) 1.295, SGR (33.607 km) 2.409, T41 (35.177 km) 2.692, AXH (35.344 km) 1.415, +1 more
noaa_gfs	gh_500	gfs:29.75,-95.50 (12.98 km) 5890, gfs:29.75,-95.25 (15.105 km) 5890, gfs:30.00,-95.50 (23.3 km) 5888, gfs:30.00,-95.25 (24.544 km) 5888, gfs:29.75,-95.75 (35.679 km) 5889, gfs:29.50,-95.50 (36.601 km) 5890, gfs:29.50,-95.25 (37.409 km) 5890, gfs:29.75,-95.00 (38.106 km) 5890, +3 more
noaa_gfs	pbl_height	gfs:29.75,-95.50 (12.98 km) 813.5, gfs:29.75,-95.25 (15.105 km) 779.2, gfs:30.00,-95.50 (23.3 km) 892.4, gfs:30.00,-95.25 (24.544 km) 759.7, gfs:29.75,-95.75 (35.679 km) 590.7, gfs:29.50,-95.50 (36.601 km) 575.6, gfs:29.50,-95.25 (37.409 km) 557.4, gfs:29.75,-95.00 (38.106 km) 554.8, +3 more
noaa_gfs	precipitable_water	gfs:29.75,-95.50 (12.98 km) 47.63, gfs:29.75,-95.25 (15.105 km) 48.29, gfs:30.00,-95.50 (23.3 km) 46.77, gfs:30.00,-95.25 (24.544 km) 47.52, gfs:29.75,-95.75 (35.679 km) 47.27, gfs:29.50,-95.50 (36.601 km) 48.28, gfs:29.50,-95.25 (37.409 km) 48.81, gfs:29.75,-95.00 (38.106 km) 49.07, +3 more
noaa_gfs	surface_pressure	gfs:29.75,-95.50 (12.98 km) 1010, gfs:29.75,-95.25 (15.105 km) 1012, gfs:30.00,-95.50 (23.3 km) 1009, gfs:30.00,-95.25 (24.544 km) 1010, gfs:29.75,-95.75 (35.679 km) 1009, gfs:29.50,-95.50 (36.601 km) 1011, gfs:29.50,-95.25 (37.409 km) 1012, gfs:29.75,-95.00 (38.106 km) 1013, +3 more
noaa_gfs	t_850	gfs:29.75,-95.50 (12.98 km) 18.13, gfs:29.75,-95.25 (15.105 km) 18.26, gfs:30.00,-95.50 (23.3 km) 18.15, gfs:30.00,-95.25 (24.544 km) 18.26, gfs:29.75,-95.75 (35.679 km) 18.08, gfs:29.50,-95.50 (36.601 km) 18.17, gfs:29.50,-95.25 (37.409 km) 18.21, gfs:29.75,-95.00 (38.106 km) 18.2, +3 more
noaa_gfs	u_10m	gfs:29.75,-95.50 (12.98 km) -1.314, gfs:29.75,-95.25 (15.105 km) -1.613, gfs:30.00,-95.50 (23.3 km) -1.315, gfs:30.00,-95.25 (24.544 km) -1.728, gfs:29.75,-95.75 (35.679 km) -1.142, gfs:29.50,-95.50 (36.601 km) -1.75, gfs:29.50,-95.25 (37.409 km) -1.721, gfs:29.75,-95.00 (38.106 km) -1.611, +3 more
noaa_gfs	v_10m	gfs:29.75,-95.50 (12.98 km) 1.349, gfs:29.75,-95.25 (15.105 km) 1.154, gfs:30.00,-95.50 (23.3 km) 0.9796, gfs:30.00,-95.25 (24.544 km) 1.035, gfs:29.75,-95.75 (35.679 km) 1.373, gfs:29.50,-95.50 (36.601 km) 1.193, gfs:29.50,-95.25 (37.409 km) 1.277, gfs:29.75,-95.00 (38.106 km) 1.425, +3 more
openaq	ozone	319 (10.025 km) 0.02166, 2046 (10.112 km) 0.01843, 310 (11.331 km) 0.02573, 3378 (15.437 km) 0.01784, 325 (16.833 km) 0.01946, 2039 (16.937 km) 0.02149, 311 (17.015 km) 0.02247, 320 (24.007 km) 0.01712, +5 more
openaq	pm10	13380082 (15.437 km) 34.76, 271 (29.74 km) 21.1
openaq	pm25	2071327 (0.02 km) 9.935, 10107877 (1.373 km) 6.827, 8967193 (4.942 km) 6.612, 15596557 (6.193 km) 5.764, 8967217 (7.937 km) 5.422, 15666344 (8.753 km) 2.576, 12286630 (9.769 km) 1.978, 2071331 (10.112 km) 11.07, +20 more
openweather	cloud_cover	grid:center (6.296 km) 44.99, grid:north (19.05 km) 55.82, grid:west (23.99 km) 48.12, grid:east (27.371 km) 44.39, grid:south (31.027 km) 33.22
openweather	humidity	grid:center (6.296 km) 77.99, grid:north (19.05 km) 79.01, grid:west (23.99 km) 78.08, grid:east (27.371 km) 82.57, grid:south (31.027 km) 76.6
openweather	precipitation	grid:center (6.296 km) 0.1393, grid:north (19.05 km) 0.1542, grid:west (23.99 km) 0.1617, grid:east (27.371 km) 0.04167, grid:south (31.027 km) 0.02014
openweather	pressure	grid:center (6.296 km) 1013, grid:north (19.05 km) 1014, grid:west (23.99 km) 1013, grid:east (27.371 km) 1014, grid:south (31.027 km) 1013
openweather	temperature	grid:center (6.296 km) 28.67, grid:north (19.05 km) 28.17, grid:west (23.99 km) 28.35, grid:east (27.371 km) 28.04, grid:south (31.027 km) 28.16
openweather	wind_direction	grid:center (6.296 km) 123.9, grid:north (19.05 km) 123.8, grid:west (23.99 km) 151, grid:east (27.371 km) 99.46, grid:south (31.027 km) 120.1

Source	Metric	Values
openweather	wind_speed	grid:center (6.296 km) 3.088, grid:north (19.05 km) 2.366, grid:west (23.99 km) 1.957, grid:east (27.371 km) 2.367, grid:south (31.027 km) 2.161
purpleair	pm25	223813 (2.634 km) 6.24, 246011 (2.666 km) 5.139, 302037 (3.437 km) 7.188, 155367 (3.772 km) 6.9, 127451 (4.843 km) 1.491, 186303 (5.079 km) 6.775, 27821 (5.147 km) 0.3053, 296101 (5.382 km) 6.589, +57 more
tceq	co	482011052 (0.0 km) 0.3264, 482011035 (15.44 km) 0.1648, 482011039 (29.756 km) 0.1375
tceq	no2	482011052 (0.0 km) 9.394, 482010047 (10.018 km) 5.825, 482010024 (11.323 km) 0.2182, 482011066 (14.45 km) 5.29, 482011035 (15.44 km) 7.788, 482010416 (16.841 km) 4.974, 482010055 (17.021 km) 3.6, 482010026 (25.342 km) 8.151, +5 more
tceq	so2	482011035 (15.44 km) -0.07123, 482010051 (22.762 km) -0.2917, 482011039 (29.756 km) 0.197

Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly



Event-nearest wind vectors from stored values (arrow points downwind)



Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). The labeling guide that comes with this packet has the definitions and marking rules. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Claim 1 of 35

A broader urban-industrial NO_x and photochemical environment is supported because the Sentinel-5P NO₂ column near the site was elevated to 4.384e-05 mol/m² around 20:07 UTC and nearby surface ozone reached 0.052 ppm at 20:00 UTC under hot afternoon conditions near 34 C.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 2 of 35

The air quality anomaly in Houston, Texas is caused by high levels of SO₂ particles.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 3 of 35

The air quality anomaly in Houston, Texas is characterized by elevated levels of PM_{2.5} and NO₂.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 4 of 35

The wind direction and speed were not significantly different from the average values for this time of year.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 5 of 35

The anomaly time and location coincide with a high local Sentinel-5P tropospheric NO2 column of 4.384018830023706e-05 mol/m^2 at 2026-06-01T20:07:14+00:00, which is above the 2026-05-31 to 2026-06-03 mean of the listed nearby granules.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 6 of 35

The air quality anomaly is related to PM2.5 levels.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 7 of 35

The anomaly occurred on June 1st at around 20:00 UTC, with a nearest sensor located approximately 2.634 km away from the anomaly.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 8 of 35

Light to moderate easterly and southeasterly surface winds facilitated the localized transport and accumulation of nitrogen oxides near the monitoring location.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 9 of 35

The air quality anomaly is caused by a nearby industrial facility releasing excessive amounts of particulate matter and volatile organic compounds.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 10 of 35

Localized combustion activities, such as industrial emissions or heavy vehicular traffic near TCEQ monitor 482011052, served as the primary physical source of the acute surface NO₂ concentration spike.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 11 of 35

Sentinel-5P satellite observations detected an elevated nitrogen dioxide column density of 4.38e-5 mol/m² over Houston near the anomaly site at 20:07 UTC on June 1, 2026.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 12 of 35

The anomaly is caused by a combination of factors including high temperatures, humidity, and stagnant atmospheric conditions, leading to the accumulation of pollutants in the air.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 13 of 35

Easterly to east-southeasterly winds of about 3 to 4 m/s likely advected NO_x-rich emissions from the Houston ship channel or other source areas east of the monitor toward the anomaly location at the event time.

Calm-wind flag: wind direction unstable under calm conditions. noaa_gfs: event wind 1.0267283585858855 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave 0.33535540855344603 m/s; n=132 wind readings). Cutoff floor: confirmed 2026-07-24.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 14 of 35

A TCEQ ground monitoring station located at (29.814, -95.388) in Houston recorded a nitrogen dioxide (NO₂) concentration of 33.5 ppb on June 1, 2026, at 20:00 UTC, exceeding the expected value of 8.84 ppb with a z-score of 8.05.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 15 of 35

The TCEQ monitor at the anomaly location measured 33.5 ppb NO₂ at 2026-06-01T20:00:00+00:00, which is a severe positive excursion relative to the expected 8.84 ppb baseline.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 16 of 35

Ground-level observations from TCEQ and spaceborne column measurements from Sentinel-5P both recorded coincident elevated nitrogen dioxide concentrations in Houston around 20:00 UTC on June 1, 2026.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 17 of 35

The temperature on June 1, 2026 was above average, with a mean value of 33.89°C.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 18 of 35

The air quality anomaly in Houston, Texas is caused by high levels of PM_{2.5} particles.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 19 of 35

Strong daytime photochemical production and accumulation over the Houston urban-industrial air mass likely increased the regional NO₂ background on 2026-06-01 because satellite NO₂ column and HCHO were elevated near the event time and surface ozone was moderately high, even though sulfur dioxide and particulate matter did not indicate a major smoke or sulfur-rich plume.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 20 of 35

A severe thunderstorm or tornado in the Houston area has stirred up dust and debris, contributing to the air quality anomaly.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 21 of 35

A severe nitrogen dioxide anomaly reaching 33.5 ppb was recorded by a TCEQ surface monitor in Houston at 20:00 UTC on June 1, 2026.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 22 of 35

A sulfur-rich industrial upset or regional smoke episode is not strongly supported because SO₂ remained near zero at the nearest TCEQ monitor, Sentinel-5P SO₂ was low near the event time, CO was only moderate at 0.4 ppm, and PM_{2.5} remained only modestly elevated near about 10 to 12 ug/m³.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 23 of 35

Meteorology near the anomaly time was consistent with transport of emissions from sources east to southeast of the monitor because nearby observations showed easterly winds near 80 to 100 degrees, wind speeds near 3.1 to 3.6 m/s, hot temperatures near 33.9 to 34.0 C, and a deep afternoon boundary layer near 1864 m.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 24 of 35

Global Forecast System boundary layer height estimates exceeding 1700 meters at 18:00 UTC indicate active vertical mixing, suggesting that high surface NO₂ concentrations resulted from intense localized source activity rather than low boundary layer stagnation.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 25 of 35

The PM2.5 levels are elevated at 9.6 ug/m3, which is above the mean of 6.0837 ug/m3.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 26 of 35

Sentinel-5P satellite observations retrieved an elevated NO2 column density of approximately 4.38 multiplied by 10 to the power of negative 5 mol/m² near the anomaly location at 20:07 UTC on June 1, 2026.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 27 of 35

At 2026-06-01T20:00:00+00:00 near the anomaly location, TCEQ CO was 0.4 ppm, TCEQ SO2 was 0.0 ppb at the nearest listed SO2 monitor 15.44 km away, OpenAQ ozone was 0.052 ppm about 10.025 km away, and nearby PM2.5 was 12.4 ug/m3 in OpenAQ and 9.6 ug/m3 in PurpleAir.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 28 of 35

ASOS surface observations and NOAA GFS weather models documented moderate easterly surface winds and a deep planetary boundary layer exceeding 1,800 meters during the anomaly period.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 29 of 35

The TCEQ NO2 monitor at 29.814, -95.388 in Houston measured 33.5 ppb at 2026-06-01T20:00:00+00:00, compared with an expected value of 8.844444444444447 ppb, corresponding to a z-score of 8.051649034639064 and a severe anomaly classification.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 30 of 35

A fresh local combustion plume from mobile traffic or nearby industrial combustion sources is a likely cause of the 33.5 ppb surface NO2 spike because the anomalous monitor also measured elevated CO at the same time, while NO2 at that monitor was far above its own typical level.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 31 of 35

A fresh local combustion plume is supported because the TCEQ monitor measured an extreme surface NO2 concentration of 33.5 ppb at 20:00 UTC while nearby winds were easterly to east-southeasterly at about 3 to 4 m/s, a pattern consistent with transport from emission sources located east of the monitor rather than stagnant accumulation.

Calm-wind flag: wind direction unstable under calm conditions. noaa_gfs: event wind 1.026728358585855 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave 0.33535540855344603 m/s; n=132 wind readings). Cutoff floor: confirmed 2026-07-24.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 32 of 35

The physical signature favors a fresh local NO_x plume over smoke or a sulfur-rich release because NO₂ was elevated at the surface and in the satellite column, while ozone was only moderately elevated and CO, PM_{2.5}, PM₁₀, and SO₂ lacked comparable extreme enhancements near the anomaly time.

Calm-wind flag: wind direction unstable under calm conditions. noaa_gfs: event wind 1.0267283585858855 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave 0.33535540855344603 m/s; n=132 wind readings). Cutoff floor: confirmed 2026-07-24.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 33 of 35

High daytime ambient temperatures reaching 34 °C accelerated photochemical oxidation of NO into NO₂, co-occurring with elevated regional ozone concentrations.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 34 of 35

The air quality anomaly in Houston, Texas is caused by high levels of NO₂ particles.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 35 of 35

Surface wind vector measurements from ASOS and OpenWeather showing winds from the east-southeast at 3.1 to 3.6 m/s support transport of precursor emissions from industrial zones east of the monitor.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.
